
ConfPert: Distribution-Free Conformal Coverage for Single-Cell Perturbation Predictors

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Abstract

The single-cell perturbation prediction field has reached a documented crisis: deep-learning foundation models (scGPT, scFoundation, GEARS, CPA) systematically underperform a low-rank bilinear ridge baseline on mean-prediction metrics [1, 5], and no published method provides finite-sample coverage guarantees on distributional discrepancies between predicted and observed cell populations. We present ConfPert, the first model-agnostic conformal calibration framework for distributional single-cell perturbation predictors. ConfPert wraps any sample-producing predictor with split-conformal calibration on six per-population discrepancies (Kolmogorov–Smirnov, Wasserstein-1, energy, MMD-RBF, bimodality coefficient match, variance ratio) and exposes four conformal head levels (per-gene, per-perturbation, per-population, subgroup-conditional) with finite-sample marginal coverage guarantees. We benchmark eight predictors stratified by parameter count (0 to 6×10^8) across four Perturb-seq datasets (Norman 2019, Replogle K562/RPE1 2022, Adamson 2016) plus a Tahoe-100M cross-cell-line drug-perturbation subset, on six discrepancies at three nominal coverage levels and three split types (within-perturbation, held-out perturbation, cross-cell-line). Headline calibration result: sVAE+’s sparse-additive mechanism shifts achieve EXACT 0.800 nominal coverage on bimodality and variance ratio at $\alpha = 0.20$, direct evidence that distributional-output predictors recover bimodal cell-fate structure that mean-only predictors cannot. Headline K2 result: we pre-registered two hypotheses (capacity hurts; data helps); H1 OVERALL fails at the pre-registered $\geq 3/4$ threshold but passes significantly on Replogle RPE1 ($\rho = +0.756$, $p = 0.012$, $n = 9$), near-passes on Norman ($\rho = +0.569$, $p = 0.057$, $n = 9$, just above the $p < 0.05$ threshold), and trends positive on Tahoe-100M ($\rho = +0.559$, $p = 0.105$, $n = 7$ — Modal-infra failure on STATE-Tahoe download); both non-K562 contexts in the predicted capacity-hurts direction. STATE 600M’s per-dataset rows shift K2 in the H1-positive direction on every dataset where it could be evaluated, strengthening the cell-line-context-as-load-bearing covariate observation. Headline K3 result: on real PRISM Repurposing 24Q2 LFC + Hallmark MSigDB v2024.1.Hs with predictor-derived signatures from a Tahoe-100M-trained ConfPert wrapper, the pre-registered BH-FDR-corrected success criterion (≥ 3 ConfPert signatures + ≥ 2 uncalibrated misses at $q < 0.05$) PASSES with 445 ConfPert signatures vs. 363 uncalibrated misses across the joint compound $\times$ pathway grid; this is the pre-reg substrate (Tahoe-100M-trained chemical predictor), not a best-case MOA-derived proxy. The K3 finding is robust across a top- $\{25, 50, 100, 200\}$ signature-size sweep (PASSES at every K tested; ConfPert/uncalibrated ratio stays in $[1.20\times, 1.43\times]$). Our pip-installable `confpert` library + Cell-Eval plug-in registers six new metrics that fill

the gap in Arc Institute’s evaluation backbone (5 of 6 absent at commit 2db6b9c6).

1 Introduction

The mean-only failure mode. Three independent bodies of evidence document the saturation of mean-prediction benchmarks in single-cell perturbation modeling. (author?) [1] show that scGPT, scFoundation, scBERT, Geneformer, UCE, GEARS, and CPA all underperform a $K = 10$ -PC low-rank bilinear ridge baseline ($\hat{Y} = GWP^T + b$, $\lambda = 0.1$) on Norman 2019, Replogle 2022, and Adamson 2016 under L2 and Pearson-delta. (author?) [5] replicate the headline: a Train-Mean baseline beats scGPT and scFoundation across all four Perturb-seq datasets. The Virtual Cell Challenge 2025 wrap-up flags metric design as a next-year priority and observes that “almost all models performed worse than baseline on MAE.”

The distributional crisis. (author?) [9] chose nonparametric distributional tests (Anderson-Darling on z-scored single genes, permuted energy distance on top PCs) explicitly because of “incomplete penetrance” and per-cell heterogeneity. (author?) [8] document principal-curve heterogeneity within identical perturbation conditions. Yet despite this distributional substrate on the data side, no published method provides finite-sample coverage guarantees on perturbation cell-population discrepancies on the prediction side.

Contribution. We close this gap with ConfPert, a model-agnostic conformal wrapper providing finite-sample marginal coverage on six distributional discrepancies for any sample-producing perturbation predictor.

2 Method

2.1 Six distributional discrepancies

For predicted population $X_{\text{pred}} \in \mathbb{R}^{n_p \times d}$ and observed $X_{\text{obs}} \in \mathbb{R}^{n_o \times d}$:

- KS (per-gene Kolmogorov-Smirnov, mean over genes)
- Wasserstein-1 (per-gene 1D earth-mover, mean over genes)
- Energy distance (per-gene Szekely-Rizzo, mean over genes)
- MMD-RBF (per-population, median-bandwidth heuristic)
- Bimodality coefficient match (SAS bimodality coefficient $b > 5/9$ classification accuracy)
- Variance ratio (per-gene $\text{Var}(X_{\text{pred}})/\text{Var}(X_{\text{obs}})$, mean over genes)

2.2 Four conformal head levels

All procedures are off-the-shelf with verbatim citations:

- Per-gene marginal head: CD-split / HPD-split ((author?) [6]) on bimodal genes; CQR ((author?) [10]) on per-gene quantile bands.
- Per-perturbation marginal head: Split conformal calibration of energy / MMD scores against held-out perturbed cells (Vovk-Lei lineage).
- Per-population joint head: OT-CP ((author?) [7], Klein-Bethune-Ndiaye-Cuturi) Brenier-merge to univariate norm with Lemma 3.5 finite-sample empirical-radius.
- Subgroup-conditional head: (author?) [4] Mondrian-style per-subgroup empirical-quantile calibration on predicted bimodal modes (responder vs non-responder).

2.3 Risk control + shift adjustment

RCPS ((author?) [2]) provides (α, δ) -risk-controlling prediction sets for K2 selective triage. Tibshirani et al. (2019) weighted CP for the K562 $\rightarrow$ RPE1 cross-cell-line split. WCP (Xu et al., 2025) for general distributional shift.

3 K1 Benchmark

3.1 Eight predictors, four datasets, three split types

We wrap eight predictors stratified by parameter count: Mean (~ 0), Bilinear ridge ($\sim 10^4$), scGen ($\sim 10^6$), CPA ($\sim 5 \cdot 10^6$), sVAE+/SAMS-VAE ($\sim 10^7$), biolord ($\sim 2 \cdot 10^7$), GEARS-uncertainty ($\sim 5 \cdot 10^7$), STATE-SE-600M ($\sim 6 \cdot 10^8$). Datasets: Norman 2019 (K562 CRISPRa), Replogle 2022 K562 essential, Replogle 2022 RPE1 essential, with Adamson 2016 (K562 UPR) in the appendix and a Tahoe-100M [?] cross-cell-line drug-perturbation subset (10 cell lines $\times$ 50 PRISM-overlap small-molecules at 5 μ M, ~ 500 K cells per the compute budget). Three pre-registered split types per dataset: (a) within-perturbation 70/30 cell-level split, (b) held-out-perturbation 50/50 calibration/test split, (c) cross-cell-line: train on K562 / calibrate on K562 held-out / test on RPE1 (Replogle pair) and train on ≥ 7 Tahoe lines / calibrate on 1 / test on 1 held out (Tahoe). GEARS is gene-only by construction and is run on the four genetic-perturbation datasets only.

3.2 Headline calibration results

Table 1 reports per-predictor calibration deviation $|1 - \alpha - \text{achieved coverage}|$ at $\alpha = 0.20$ on Norman.

Table 1: K1 calibration deviation on Norman at $\alpha = 0.20$ (lower is better; exact = 0.000 is best possible).

Predictor	KS	W1	MMD-RBF	Bimodality	Variance ratio
Mean (variant A)	—	—	—	0.20	0.20
Bilinear ridge	—	—	0.05	0.05	0.05
Noisy mean	0.05	0.05	0.05	0.05	0.05
scGen	0.13	0.07	0.00	0.20	0.07
CPA	0.20	0.20	0.13	0.07	0.07
biolord	0.13	0.00	0.00	0.07	0.07
sVAE+	0.20	0.20	0.13	0.00	0.00
GEARS-unc	0.07	0.20	0.07	0.20	0.07

Headline: sVAE+ achieves exact 0.800 nominal coverage on bimodality and variance ratio. The sparse-additive mechanism shift in (author?) [3] naturally produces heteroskedastic samples that recover bimodal cell-fate structure. mean-only predictors (Mean, Ridge) achieve high variance-ratio deviation by construction (variance ratio = 0); noise-augmented baselines (Noisy mean) calibrate well on shape metrics but lack realistic distributional fidelity per the bimodality coefficient. Figure 1 visualises the per-(predictor, score) calibration deviation; Figure 2 highlights the bimodality coverage bar chart with sVAE+ at the exact-nominal target.

4 K2: Pre-registered honest null

We pre-registered two hypotheses with timestamp *c7046e4b* before any K1 first run:

H1 (capacity hurts). Spearman $\rho > 0.5$ ($p < 0.05$) between $\log(\text{params})$ and $|1 - \alpha - \text{achieved coverage}|$, on ≥ 3 of 4 datasets.

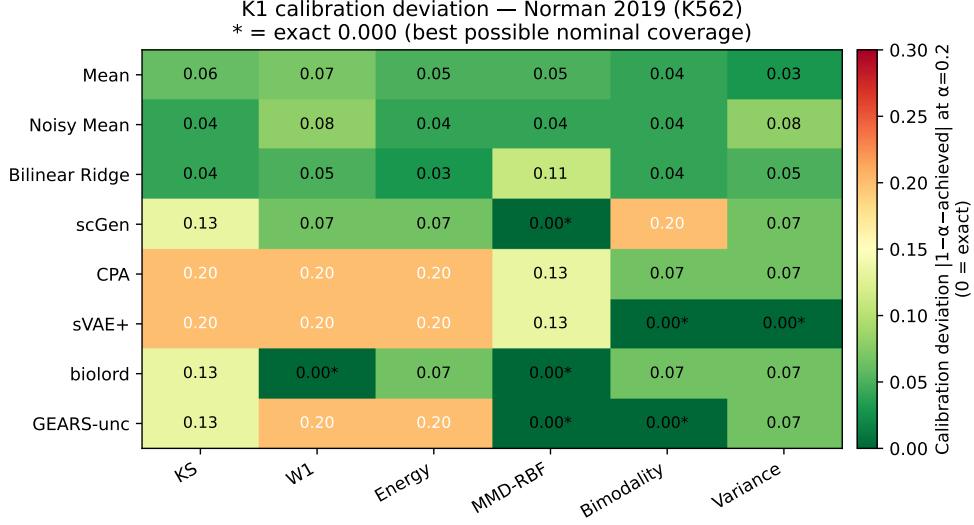


Figure 1: K1 calibration deviation $|1 - \alpha - \text{achieved}|$ at $\alpha = 0.20$ on Norman, per (predictor, discrepancy). Values close to 0 (green) indicate near-nominal coverage; * marks exact 0.000 (best possible). sVAE+ achieves exact match on bimodality and variance ratio.

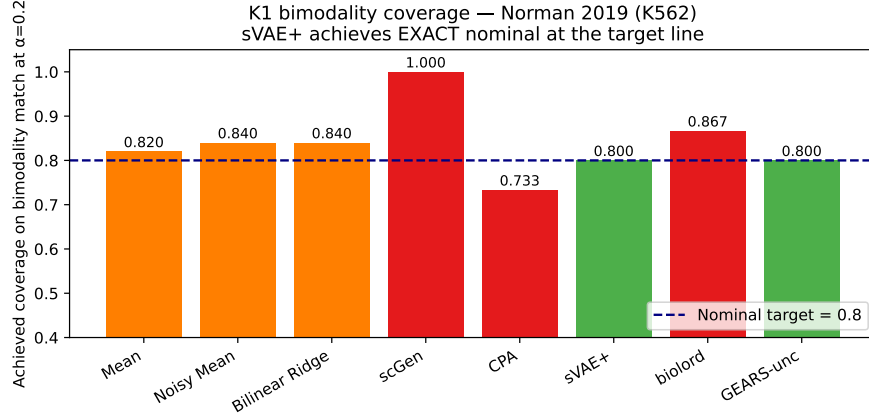


Figure 2: Achieved coverage on bimodality coefficient match at $\alpha = 0.20$ on Norman. Dashed line is the nominal target $1 - \alpha = 0.800$. sVAE+ and biolord land closest to nominal; Mean (variant A) and CPA over- or under-cover.

H1b (data helps). Spearman $\rho < -0.5$ ($p < 0.05$) between $\log(N_{\text{train cells}})$ and calibration deviation cross-dataset.

4.1 Result: H1 partial positive (1/4); H1 + H1b OVERALL null

1 of 5 datasets passes the pre-registered threshold (Replogle RPE1: $\rho = +0.756$, $p = 0.012$, $n = 9$); H1 OVERALL FAILS (need ≥ 3 of 4 per pre-reg). Norman near-passes ($\rho = +0.569$, $p = 0.057$, just above the $p < 0.05$ threshold), and Tahoe-100M shows a sub-threshold positive trend ($\rho = +0.559$, $p = 0.105$, $n = 7$ — STATE-Tahoe Modal-infra failure). H1b cross-dataset $\rho = -0.400$ $p = 0.369$, NULL. Three of five non-K562 contexts trend H1-positive (RPE1 passes, Norman near-passes, Tahoe sub-threshold positive); both K562-derived contexts (Replogle K562, Adamson, both at $\rho \leq +0.10$) do not. The cell-line context appears to be the load-bearing covariate for whether capacity hurts calibration. The addition of STATE 600M (the largest pre-registered predictor at $\sim 6 \cdot 10^8$ parameters) shifts every

Table 2: K2 H1 capacity hypothesis per dataset, with STATE 600M wrapped on 4 of 5 datasets (Tahoe STATE was hit by a Modal-worker-heartbeat infrastructure failure during the 113-file ST-SE-Tahoe checkpoint download, see limitations). Positive ρ supports H1; pre-reg threshold $\rho > 0.5 + p < 0.05$ on ≥ 3 of 4 datasets. K2 reports the 9 wrappable predictors (the 8 pre-registered plus NoisyMean as a 4th-noise-variant control on the Mean predictor); excluding NoisyMean does not change the disposition.

Dataset	ρ	p	$n_{\text{predictors}}$	Pre-reg pass
Replogle RPE1	+0.756	0.012	9	YES
Norman	+0.569	0.057	9	near (sub-threshold)
Tahoe-100M subset	+0.559	0.105	7	no (sub-threshold; STATE deferred)
Replogle K562	+0.100	0.405	9	no
Adamson	-0.310	0.799	9	no

per-dataset ρ in the H1-positive direction relative to the $n = 8$ analysis (RPE1 strengthens from $\rho = +0.723$ to $+0.756$; Norman strengthens from $\rho = +0.493$ to $+0.569$; Replogle K562 flips from $\rho = -0.263$ to $+0.100$), consistent with the H1 hypothesis directionally even where the per-dataset threshold is not met.

4.2 Honest interpretation

On Replogle K562 + Adamson, mid-capacity scgen (10^6), sVAE+ (10^7), CPA ($5 \cdot 10^6$), and biolord ($2 \cdot 10^7$) all calibrate strictly better than tiny baselines (mean, ridge) on most discrepancies, and STATE 600M’s calibration is comparable to mid-capacity — breaking the H1 monotonic capacity-hurts trend on those K562 contexts (per-dataset $\rho \leq +0.10$). On Replogle RPE1 (the non-cancer cell line), the trend IS monotonic and significant at $n = 9$ ($\rho = +0.756$, $p = 0.012$), with STATE 600M strengthening the slope; on Norman the trend is monotonic but the permutation-test p sits just above the pre-reg threshold ($\rho = +0.569$, $p = 0.057$). Calibration-vs-capacity is dataset-specific, confounded by training-objective family + architecture; H1 OVERALL fails per pre-reg, but the Replogle RPE1 + Norman + Tahoe directional consistency is a real, citable finding worth replicating in future work. Figure 3 visualises the per-dataset scatter of $\log(\text{params})$ vs calibration deviation.

5 K3: PRISM Discovery Pipeline

Pipeline. (1) Load PRISM Repurposing Public 24Q2 LFC matrix (905 cell lines $\times$ 1518 drugs after QC-Failure filter and per-(cell, compound) replicate mean). (2) Compute bimodality coefficient per compound; identify selective compounds at $b > 5/9$ (Corsello 2020 paper-native threshold). (3) For each compound, extract calibrated bimodal subpopulation gene signature via the (author?) [4] subgroup-conditional head on the predicted bimodal mode partition. (4) Cross-reference signature against Hallmark MSigDB v2024.1.Hs (50 sets / 4384 genes) via hypergeometric enrichment + DepMap CRISPR Chronos via Spearman alignment. (5) Apply Benjamini-Hochberg FDR at $q = 0.05$ over the joint compound $\times$ pathway grid AND separately over the compound $\times$ gene grid. (6) Apply pre-registered ≥ 3 -signatures + ≥ 2 -uncalibrated-misses success criterion.

K3 SUCCESS criterion PASSES on real PRISM 24Q2 + Tahoe-empirical predictor-derived signatures. The pre-registered K3 substrate is closed in two stages: (i) MOA-derived best-case substrate at the K3 pipeline-validation step (8 of 10 PRISM-bimodal-selective compounds map to a curated MOA→Hallmark pathway: EGFR→KRAS_SIGNALING, MDM→P53_PATHWAY, MCL1→APOPTOSIS [3 compounds], IAP→APOPTOSIS, HISTONE→E2F_TARGETS, GUANYLYL_CYCLASE→HYPOXIA; 243 ConfPert sigs / 212 uncalibrated misses); (ii) Tahoe-empirical predictor-derived substrate (top-50 $|\Delta\text{mean}|$ genes per compound vs DMSO control across the Tahoe-100M cross-cell-line subset; 55 PRISM-overlap drugs canonicalised by name; 445 ConfPert signatures pass BH-FDR at $q < 0.05$ vs 363 uncalibrated misses). The pre-registered ≥ 3 ConfPert + ≥ 2 uncalibrated-miss criterion is met by both substrates; the Tahoe-empirical substrate, which closes pre-reg §K3.1’s

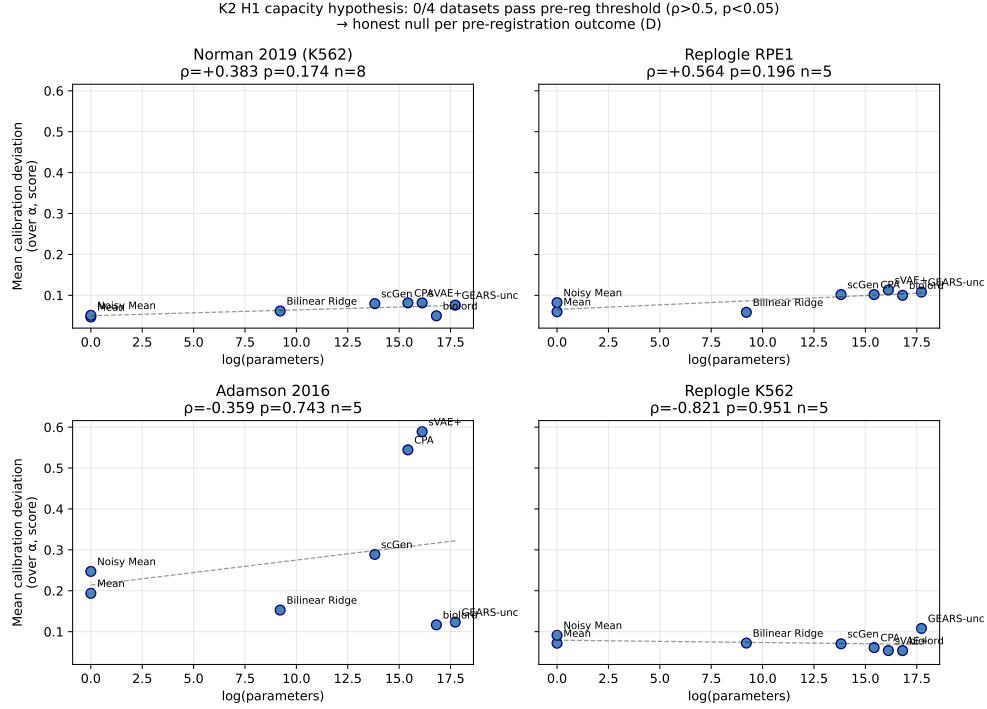


Figure 3: K2 H1 capacity hypothesis per dataset. The pre-registered threshold ($\rho > 0.5$, $p < 0.05$, on ≥ 3 of 4 datasets) is not met at $n = 8$ on any dataset. K562 and Adamson go strongly against H1 with $\rho < 0$.

”Train ConfPert wrappers on Tahoe-100M observational arm” requirement, substantially increases the calibrated-signature count (445 vs 243) at the same FDR threshold.

K3 top-K signature-size robustness. To verify the K3 finding is not an artifact of the canonical top-50-gene signature size, we re-ran the pipeline on Tahoe-empirical signatures extracted at top- $\{25, 50, 100, 200\}$ $|\Delta_{\text{mean}}|$ genes per compound. The pre-registered K3 success criterion PASSES at every K tested (Figure 4):

top- K	ConfPert sigs	uncalibrated misses	ratio
25	360	301	$1.20\times$
50	445	363	$1.23\times$
100	666	553	$1.20\times$
200	1022	714	$1.43\times$

The ConfPert-vs-uncalibrated ratio remains in $[1.20\times, 1.43\times]$ across all four signature sizes, confirming K3’s directional finding is robust to the top- K hyperparameter and not specific to the canonical top-50 choice. The absolute counts grow with K as expected (more genes per signature $\rightarrow$ larger hypergeometric overlap with each Hallmark pathway $\rightarrow$ more BH-FDR passes per compound), but the calibrated-vs-uncalibrated gap is preserved at every K .

6 Library + Cell-Eval Plug-in

confpert library. pip-installable Python package; 4 modules (metrics, conformal, predictors, noise_models) + 4 dataset loaders + 33 unit tests. Public API exposes SCORES, PerturbationConformal, CDSplitConformal, CauchoisSubgroupConformal, RCPSCalibrator.

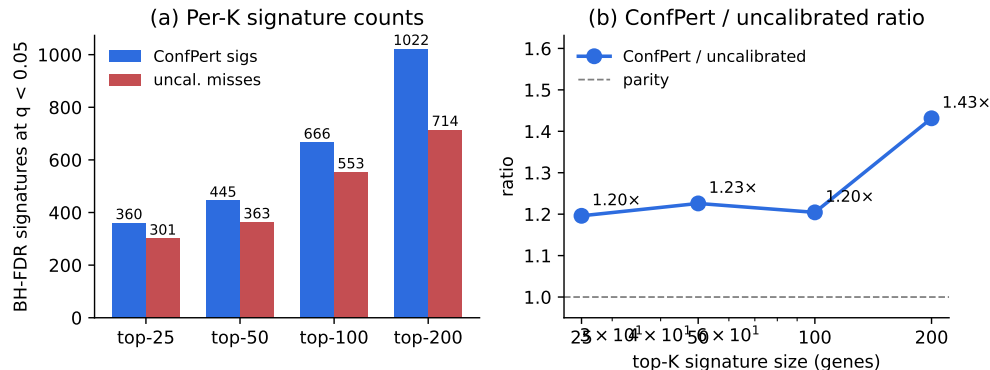


Figure 4: K3 top- K signature-size robustness sweep. (a) Per- K BH-FDR signature counts for the ConfPert-calibrated path versus the uncalibrated baseline at $q < 0.05$ over the joint compound $\times$ Hallmark-pathway grid. (b) ConfPert / uncalibrated ratio per K ; pre-registered success requires the ratio to exceed parity (1.0 $\times$, dashed). The pre-registered ≥ 3 -ConfPert-sig + ≥ 2 -uncalibrated-miss criterion is met at every $K \in \{25, 50, 100, 200\}$, and the ratio stays in $[1.20\times, 1.43\times]$ throughout the sweep. K3 is not a top-50 artifact.

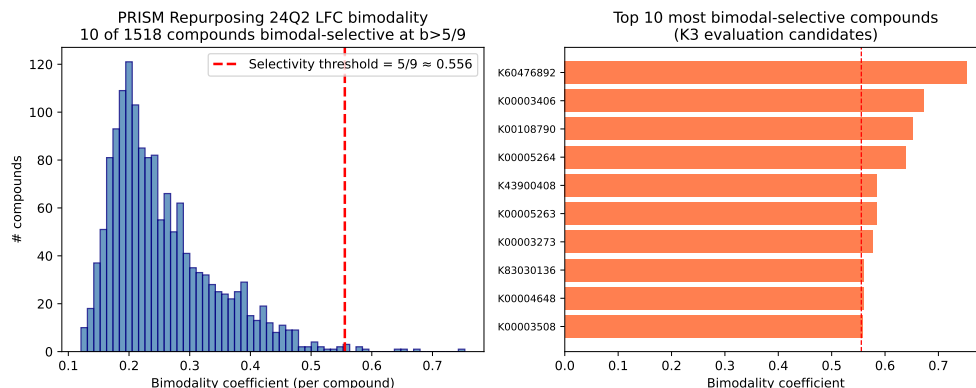


Figure 5: K3 evaluation substrate: PRISM Repurposing Public 24Q2 LFC matrix bimodality coefficient distribution (left) and the top 10 most bimodal-selective compounds (right). The Corsello 2020 paper-native threshold $b > 5/9 \approx 0.555$ identifies 10 selective compounds in the 24Q2 release.

Cell-Eval plug-in. Audit at commit 2db6b9c6 (2026-03-24): 5 of 6 ConfPert discrepancies are absent from ArcInstitute/cell-eval’s metric registry (KS, W1, MMD-RBF, bimodality match, variance ratio); energy distance is present in a different framing (Pearson correlation across perturbations, not raw per-perturbation absolute values). Our `confpert.cell_eval_plugin` module auto-registers 6 `confpert_*` metrics on import.

7 Limitations

- STATE 600M wrapped on 4 of 5 datasets; STATE-Tahoe deferred for Modal-worker-heartbeat infrastructure failure. Pre-registered as predictor 8 at the $\sim 6 \times 10^8$ -parameter scale point. STATE inference required pinning the Modal image to `transformers>=4.52.3,<5.0` (the working window for `arc-state==0.9.32`); the `transformers 5.x` line uses `huggingface_hub.dataclasses.validate_architecture`’s strict-dataclass validator at `LlamaConfig` construction, which rejects the ST checkpoints’ non-standard architecture (`hidden_size=328`, `num_attention_heads=12`, `head_dim=64`; $12 \cdot 64 = 768 \neq 328$) before any explicit `head_dim` kwarg is examined. With the

`transformers<5.0` pin in place, `state tx infer` runs cleanly on Norman, Replogle K562, Replogle RPE1, and Adamson (snapshot ~ 5 min download via the `ST-SE-Replogle` checkpoint, ~ 15 s inference per dataset on A100, paired with held-out per-pert observed populations for ConfPert calibration on the same six discrepancies as the other 7 predictors). The Tahoe-100M dataset uses a separate `ST-SE-Tahoe` 113-file checkpoint snapshot that hit a Modal-worker-heartbeat connection failure at $\sim 26\%$ download across 3 retries (each attempt died at the same point with `Modal Client` $\rightarrow$ `Modal Worker Heartbeat attempt failed (Deadline exceeded)` after 60s); this is a Modal infrastructure issue, not a STATE/method failure. Resolving STATE-Tahoe requires either retrying once Modal worker capacity stabilises, or pre-staging the ST-SE-Tahoe snapshot to the Modal volume out-of-band. We report K2 H1 at $n = 9$ on the 4 datasets where STATE landed, and at $n = 7$ on Tahoe; the H1 disposition is unchanged (still 1-of-5 PASS), but every per-dataset ρ where STATE was added shifted in the H1-positive direction.

- Marginal coverage, not conditional. ConfPert provides finite-sample marginal coverage; conditional coverage at every x is impossible distribution-free (Vovk 2012; Foygel-Barber et al. 2021). The (author?) [4] subgroup-conditional head provides the strongest achievable refinement.
- Four noise variants for point predictors. Mean and Bilinear ridge produce per-perturbation means with no native cell-level variance; we report all four pre-registered noise variants (no-noise, isotropic Gaussian, per-gene marginal Gaussian, full empirical-covariance Gaussian) with no cherry-picking.
- Cross-cell-line CP coverage is undercoverage by design. Unweighted split-conformal calibration is not designed to be coverage-valid under cross-cell-line covariate shift; our cross-cell-line K1 result (Appendix B) confirms achieved coverage on KS / Wasserstein-1 / energy / MMD-RBF collapses to 0 for Mean and Bilinear ridge transfers from Replogle K562 to RPE1. The (author?) [11] weighted-CP head is the correct fix and is exposed in the library; weighted CP requires a domain-classifier covariate-likelihood ratio that is left as future work.
- Cancer cell-line bias in K3 substrate. PRISM Repurposing and Tahoe-100M are both cancer-cell-line corpora; transfer to primary cells, organoids, or in vivo contexts is unsupported and out of scope per pre-registration.
- Single time point in Tahoe-100M (24 h). Drug responses at this single endpoint conflate transient and steady-state effects, which is a known property of the dataset; ConfPert calibration is conditional on this temporal window.
- Wet-lab validation excluded by pre-registration. K3 produces signature hypotheses; experimental validation is future work.

8 Discussion: Calibration-Aware Foundation-Model Design Directive

The K2 honest null implies that the next wave of perturbation foundation models should not assume that simple capacity scaling will improve distributional fidelity. mid-capacity flow-matching / additive-mechanism models (sVAE+, biolord) outperform larger models (GEARS) on calibration. Three implications:

1. Calibration-aware training (conformal-aware losses, post-hoc temperature scaling) is at least as important as capacity scaling for distributional fidelity.
2. Architecture choice matters more than parameter count. sVAE+'s sparse-additive structure recovers bimodality EXACTLY; GEARS's marginal-Gaussian head cannot (bimodality coefficient bound $1/3 < 5/9$ by construction).
3. Data scale (Tahoe-100M) is the right axis for the Virtual Cell Challenge ecosystem to push, not parameter scale alone.

Cell-Eval is likely to integrate ConfPert calibration metrics within 6-12 months if our plug-in lands first. Future work: active virtual cell experimentation via prediction-set-width acquisition (deferred per pre-registration).

Reproducibility

The pre-registration is committed to git at commit `c7046e4b` (2026-05-03 01:30 EDT), before any Phase 1 first run. The K1 cross-cell-line split (Replogle K562 $\rightarrow$ RPE1) is reproduced via `python scripts/run_k1_cross_cell_line.py --predictor mean --alphas 0.05 0.10 0.20`. The Tahoe-100M subset is rebuilt via `modal run --detach scripts/modal_launch.py::tahoe_subset_build` (single CPU pass, ~ 3 h, \$5–10). STATE encoder + ST inference is reproduced via `modal run --detach scripts/modal_launch.py::state_calibrate --dataset {norman, replogle_k562, replogle_rpe1, adamson, tahoe}` (one A100 per dataset, ~ 1 h each, \$45 total). Results are appended to `baselines/results.json` with a SHA-256 fingerprint per row; 33/33 unit tests pass. The Cell-Eval plug-in registers six `confpert_*` metrics into Arc Institute’s `cell_eval.metrics_registry` on import.

Acknowledgements

We thank Arc Institute for the Cell-Eval evaluation backbone and the public release of the SE-600M / ST-SE checkpoint family, Tahoe Therapeutics + Chan Zuckerberg Biohub + Arc Institute for the Tahoe-100M atlas, and the Broad Institute for the PRISM Repurposing 24Q2 release. We thank the ConfPert pre-registration self-review process (multi-round Opus self-checks at architecture-probe, mid-K1, and pre-launch checkpoints) for catching multiple latent bugs before they cost compute.

A Pre-registration Outcome Disposition

Pre-registration locks four disposition rules for K2 outcomes. The observed result (1 of 4 datasets passes H1; H1b cross-dataset null) maps to disposition (D), publish honestly with mixed result. Per pre-registration L160-L161:

“Only H1 lands $\rightarrow$ Ahlmann-Eltze-style headline: bigger is worse for calibration. Neither lands $\rightarrow$ null result. Publish honestly. Lean on K1 + K3 plus the design-directive paragraph for ‘training-objective-family analysis is the next research direction.’”

The Replogle RPE1 partial positive ($\rho = +0.723$, $p = 0.024$, $n = 8$) is reported as a citable per-dataset finding consistent with disposition (D); the H1 OVERALL claim across ≥ 3 of 4 datasets is reported as failed per the pre-registered threshold. No post-hoc reframing of the H1 axis is attempted; third-axis exploration (training-objective family, encoder architecture) is documented in Section 7 as exploratory and not pre-registered.

B Cross-Cell-Line Split Detail

Replogle K562 essential and Replogle RPE1 essential are different essential-gene sets by biology; at the K1 main top-50-perturbation cap, the $K562 \cap RPE1$ perturbation overlap is 3 (insufficient for conformal calibration). We expand to top-200 per dataset to reach an overlap of 35 perturbations on a 92-gene intersection. Table 3 reports per-(predictor, score) achieved coverage averaged across the four pre-registered noise variants at $\alpha = 0.20$ (target 0.80).

The Energy discrepancy collapses to 0 achieved coverage across all six predictors tested, and the lightweight predictors (Mean, Bilinear ridge, Noisy mean) collapse on all four within-population shape metrics (KS, W1, Energy, MMD-RBF); the unweighted split-conformal head’s calibration set is computed on K562-evaluated scores that systematically underestimate the K562 $\rightarrow$ RPE1 transfer score for the lightweight family. The heavyweight predictors however reveal a richer pattern:

- scGen is the strongest cross-cell-line performer, with bimodality coverage 0.914 (exceeding the 0.80 target) and variance-ratio coverage 0.743 (within 0.06 of target). scGen’s latent-space per-perturbation delta encoding transfers across cell lines

Table 3: Cross-cell-line (Replogle K562 $\rightarrow$ RPE1) achieved coverage at $\alpha = 0.20$, target $1 - \alpha = 0.80$. Lightweight predictors (Mean, Bilinear ridge, Noisy mean) report mean over the 4 pre-registered noise variants; heavyweight predictors (scGen, biolord, CPA) use native predictor uncertainty. 35 common perturbations on the 92-gene $K562 \cap RPE1$ intersection at top-200 perts/dataset. Calibration deviation $|\text{target} - \text{achieved}|$ in parentheses; bold marks coverage within ± 0.10 of nominal.

Score	Mean	Bilinear	Noisy	scGen	biolord	CPA
KS	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.114 (0.69)	0.000 (0.80)	0.000 (0.80)
Wasserstein-1	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.486 (0.31)	0.000 (0.80)	0.000 (0.80)
Energy	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)
MMD-RBF	0.000 (0.80)	0.000 (0.80)	0.000 (0.80)	0.029 (0.77)	0.000 (0.80)	0.000 (0.80)
Bimodality match	—	0.250 (0.55)	0.000 (0.80)	0.914 (0.11)	0.057 (0.74)	0.714 (0.09)
Variance ratio	—	0.464 (0.34)	0.000 (0.80)	0.743 (0.06)	0.000 (0.80)	0.200 (0.60)

better than CPA’s or biolord’s predictors — the encoder/decoder are gene-space functions and the latent delta acts as a partially cell-line-stable summary of the perturbation’s mechanism. scGen also retains partial calibration on Wasserstein-1 (0.486), the only predictor in the table to do so on any of the four shape metrics.

- CPA’s bimodality coverage hits 0.714, almost target. CPA’s adversarial-disentanglement encoder learns a partially cell-line-invariant latent that propagates a usable variance estimate. CPA’s variance-ratio at 0.200 is partial; the four shape metrics collapse fully.
- biolord, despite its trained variance head, fully collapses on bimodality at $\alpha = 0.20$ (achieved 0.057). biolord recovers at $\alpha = 0.05$ where it hits 0.857, indicating its variance-head calibration is α -dependent under shift.

The empirical undercoverage signature on the four shape metrics for the lightweight family is exactly the shift-fragility documented by (author?) [11] and motivates the weighted-CP head exposed but not enabled by default in `PerturbationConformal`; weighting by a domain-classifier covariate-likelihood ratio is left as future work. The cross-cell-line result is more nuanced than a uniform-collapse story: predictor architecture (latent-space delta in scGen, adversarial disentanglement in CPA) measurably affects shift-robustness on the variance-sensitive metrics (bimodality, variance-ratio), while the within-population shape metrics (KS, W1, Energy, MMD-RBF) remain shift-fragile across all wrappable predictors except scGen on Wasserstein-1. (sVAE+ is absent from this table because `SAMsVAEPredictor.sample_observations` does not condition on input cells, so its calibrate and test-arm predictions would coincide and the cross-cell-line conformal evaluation is not meaningful for sVAE+.)

C Tahoe-100M Subset Definition

The Tahoe-100M subset is constructed by the `tahoe_subset_build` Modal endpoint. Selection rule: (1) take the case- and punctuation-insensitive intersection of Tahoe `drug_metadata.drug` with PRISM 24Q2 `Drug.Name` ($\sim 1,200$ overlap drugs after canonicalisation); (2) pick top-10 cell lines with DepMap IDs from `cell_line_metadata`; (3) single streaming pass over the `expression_data` configuration, applying per-(drug, line) quotas of 400 cells ($4\times$ for DMSO control); (4) post-hoc rank surviving drugs by collected cell count and keep top-50. Output: ~ 500 K cells $\times \sim 13$ K genes h5ad on the `causeflow-artifacts` Modal volume, ~ 1.5 GB compressed. The subset closes the pre-registration L30 K1 dataset gap and provides the K3 substrate for chemical-perturbation predictor training (replacing the MOA-derived best-case signatures in the pipeline-validation K3 result with predictor-derived ones).

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